IGCSE 0607 Practice Questions — E1.7

Indices I

Total Questions: 15 — **Total Marks:** 85 Marks

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1 **(a)** Find the value of 7^0.

..... **[1]**

(b) Find the value of $81^{\frac{1}{2}}$.

..... **[1]**

(c) Write as a single power of x.

$x^5 \times x^{-2}$

..... **[1]**

2 **(a)** Work out the value of

$125^{\frac{1}{3}}$

..... **[1]**

(b) Work out the value of

6^{-2}

..... **[1]**

(c) Simplify.

$t^3 \times t^5$

..... **[1]**

3 **(a)** Simplify.

$m^5 \times m^2$

..... **[1]**

(b) Find the value of

5^{-2}

..... **[1]**

(c) Find the value of

$49^{\frac{1}{2}}$

..... **[1]**

4 **(a)** Find the value of

3^{-2}

..... **[1]**

(b) Find the value of

$\left(\frac{8}{27}\right)^{\frac{1}{3}}$

..... **[1]**

(c) Simplify.

$x^4 \times x^{-2}$

..... **[1]**

(d) Simplify.

$(y^3)^4$

..... **[1]**

5 **(a)** Find the value of

$2^{-3} \times 2^{5}$

..... **[2]**

(b) Work out the value of

$81^{\frac{1}{2}}$

..... **[1]**

(c) Find the exact value of

64^{0}

..... **[1]**

6 **(a)** Work out the value of

$$16^{-\frac{3}{4}}$$

..... **[2]**

(b) Simplify.

$$\frac{12w^6y^3}{3w^{-2}y^5}$$

..... **[3]**

7 **(a)** Simplify fully

$(3xy^2)^3$

..... **[2]**

(b) Find the value of x when

$16^x = 2^{10}$

$x =$ ..... **[3]**

8 **(a)** Simplify.

$(4x^3y)^2$

..... **[2]**

(b) Simplify.

$$\frac{18a^6b^2}{6a^2b^5}$$

..... **[2]**

(c) Work out the value of

$27^{-\frac{2}{3}}$

..... **[2]**

9 **(a)** Find the exact value of each of the following.

(i) $64^{\frac{2}{3}}$

..... **[2]**

(ii) 2^{-4}

..... **[1]**

(b) **(i)** Simplify fully.

$$\frac{15a^6b^3}{3a^2b^7}$$

..... **[2]**

(ii) Simplify fully.

$$(4m^6n^{-2})^{\frac{1}{2}}$$

..... **[2]**

10 **(a)** Find the exact value of

$16^{-\frac{3}{4}}$

..... **[2]**

(b) Simplify fully.

$(5x^3y^{-2})^2$

..... **[2]**

(c) Solve the equation.

$2^{3x-1} \times 4^x = 32$

$x =$ ..... **[3]**

11 **(a)** Work out the exact value of

$$125^{-\frac{2}{3}}$$

..... **[3]**

(b) Simplify

$$\left(\frac{16a^8}{81b^{12}}\right)^{\frac{3}{4}}$$

..... **[3]**

12 **(a)** Simplify.

$$\left(\frac{25x^4}{16y^8}\right)^{-\frac{1}{2}}$$

..... **[3]**

(b) Solve the equation.

$$2^{3x-1} \times 4^{x+2} = 8^{x-3}$$

$x =$ ..... **[4]**

13 **(a)** Simplify fully.

$$\left(\frac{16x^8}{y^4}\right)^{-\frac{1}{2}}$$

..... **[3]**

(b) Solve the equation.

$$4^x - 2^{2x+1} + 8 = 0$$

$x =$ ..... **[4]**

14 **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{81y^{8}}\right)^{-\frac{3}{4}}$$

..... **[3]**

(b) Find the value of x when

$$8^{x} = 4^{x+3}$$

$x =$ ..... **[3]**

(c) Solve the equation.

$$5 \times 25^{k} = 125$$

$k =$ ..... **[3]**

15 **(a)** Find the value of k when $5^k = \dfrac{1}{\sqrt{125}}$.

$k =$ ..... **[3]**

(b) Simplify fully.

$$\left(\frac{16x^{12}}{81y^8}\right)^{-\frac{3}{4}}$$

..... **[3]**

(c) Solve the equation.

$$2^{2x-1} \times 4^x = 32$$

$x =$ ..... **[3]**